Govt Graduate College Burewala

Batch 22-26

Assignment: 01

Subject:	**Topic:**
Information Security	Computer Virus

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1. Computer Virus

A computer virus is a type of malicious software (**malware**) that is designed to infect a computer system, replicate itself, and spread to other devices. Similar to a **biological virus**, it attaches itself to a host file or program and becomes active when that file is executed. **Computer viruses** can damage files, steal confidential information, slow down systems, corrupt data, or completely crash a computer.

Example:

One of the most well-known examples is the **ILOVEYOU** virus (2000), which spread through infected email attachments and caused billions of dollars in damage worldwide.

A virus:

- Infects files or programs
- Replicates itself
- Spreads to other systems
- Can damage data or steal information

How a Computer Virus Works

A virus typically works in 4 stages:

1. Infection Phase

The virus attaches itself to a file or program.

2. Execution Phase

When the infected file is opened, the virus code runs.

3. Replication Phase

The virus copies itself into other files.

4. Payload Activation

The virus executes its harmful function (e.g., delete files, steal data).

Form of Computer Virus

1. Worm

◇ Definition

A **worm** is a type of malware that spreads automatically **without user interaction**.

Unlike a virus, it does **not need to attach to a file**.

◇ How It Works

- Exploits network vulnerabilities
- Self-replicates
- Spreads across computers automatically

◇ Example

WannaCry

- Spread globally in 2017
- Exploited Windows vulnerability
- Infected 200,000+ computers
- Spread automatically through networks

2. Trojan Horse

◇ Definition

A **Trojan Horse** is malware that **disguises itself as legitimate software** to trick users.

The name comes from the Greek story of the **Trojan Horse**.

◇ How It Works

- Looks like useful software (game, app, crack file)
- User installs it manually
- Hidden malicious code runs in background

◇ Example

Zeus Trojan

- Steals banking information
- Records keystrokes
- Sends data to attackers

3. Ransomware

◇ Definition

Ransomware is malware that **encrypts victim's files** and demands payment (ransom) to unlock them.

◇ How It Works

1. Infects system
2. Encrypts files
3. Shows ransom note
4. Demands payment (usually cryptocurrency)

◇ Example

Crypto Locker

- Encrypted user files
- Demanded Bitcoin payment
- Locked thousands of systems

🔎 Quick Comparison Table

Feature	Worm	Trojan	Ransomware
Needs User Action?	❌ No	☑ Yes	Sometimes
Self-Replicates?	☑ Yes	❌ No	❌ No
Disguised as Legit Software?	❌ No	☑ Yes	❌ No
Encrypts Files?	❌ No	❌ No	☑ Yes

How Viruses Are Created

Malware developers typically:

- Use programming languages like C, C++, Python, Assembly
- Study operating system internals
- Exploit system vulnerabilities
- Use obfuscation techniques to avoid antivirus detection

Ethical hackers and cybersecurity researchers study these techniques in **controlled lab environments** to:

- Develop antivirus software
- Improve system security
- Detect vulnerabilities

4. Virus Code

Example of Virus Logic (Safe Simulation)

Below is a harmless example demonstrating the concept of self-replication for educational purposes only:

python

Educational self-copy simulation (harmless example)

```
import os

import shutil

source_file = __file__

target_file = "copy_of_script.py"

if not os.path.exists(target_file):

shutil.copy(source_file, target_file)

print("File copied successfully!")

else:

print("File already exists.")
```

⚠ This program simply copies itself into another file:

It does not harm the system and is used only to demonstrate replication logic.